Company Memory for Grounded Marketing Recommendations: Separating Evidence Selection from Reader Performance

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Abstract

Company memory can improve recommendations by supplying the right evidence without changing the underlying model. We isolate this effect in 3 696 actual API calls: seven contexts, two fixed readers (GPT-6.1 Sol and Claude Sonnet 5.5), and 264 cases in 33 fictional company worlds. At a 2,048-byte evidence allowance, specialized prerequisite coverage increases grounded decision correctness from 223/264 (84.5%) to 264/264 (100%) with both readers: +15.5 percentage points, 95% world-bootstrap interval [9.8,21.2]. Positive recommendation coverage rises from 91/132 to 132/132. A strong generic structured-Markdown control, GBrain with the same coverage layer and full current context also reach 264/264; specialization is not uniquely necessary at this allowance. A retrieval-only sweep reveals its narrower advantage: at 1,024 bytes it preserves complete support in 132/132 cases versus 52/132 for generic file diversity. Separate query-blind wiki compilation and 264 exploratory QA calls on external company documents expose compression, ambiguity and reader-dependent failures. The evidence supports coverage-aware context selection as a composable capability, not universal specialized-brain superiority, commercially optimal actions or revenue lift.

1 The question: useful context, useful actions

A company memory may accurately store thousands of records yet fail to supply the few facts needed for a defensible next action. For a marketing research recommendation, those facts can span approved audience policy, an account's current situation, an offering's relevant pain and supporting proof. A recent signal alone is insufficient. Conversely, a capable reader cannot recover company-specific evidence that its context omits.

We ask whether domain-conditioned evidence selection improves *grounded action recommendations*, holding the reader fixed. This separates two interventions often bundled under a "specialized brain": changing the evidence presented to a model and changing the model's planning procedure. We measure complete evidence support before reading and action correctness afterward, using the same GPT or Claude within every comparison. The task checks the eligibility of a bounded research action; it does not identify the revenue-maximizing marketing strategy.

The contribution is an auditable controlled comparison, not a new memory architecture: (i) real GPT/Claude calls across seven memory treatments; (ii) native GBrain search and a strong structured-file reference, including a same-backend coverage control; and (iii) an exploratory raw-document versus compiled-wiki check on externally authored company material. The release retains raw responses, unsuccessful attempts, source hashes and executable scoring. Both positive effects and simple baselines that erase them matter for architecture decisions.

*AI-assisted industrial technical report. Original synthetic experiments and publicly licensed components; no customer data or human evaluation. Not peer reviewed.

Work	What it already establishes or studies	Distinction in this experiment
RAMP [1]	Client memory, marketing plans, audience filtering and reflection	Fixed planner; separate evidence support and action eligibility; two readers
CMC [2]	Resolve state before retrieval; fixed-reader context interventions	Approved policy, conflicting values, permissions and action prerequisites
WiCER [3]	Raw versus compiled wiki; compilation loss and refinement	Query-blind compilation without test-driven refinement; strict saved-answer scoring
LongMemEval [4]	Updates, temporal reasoning and abstention	Grounded action specification as well as memory availability
VehicleMemBench [5]	Domain memory connected to executable agent behavior	Marketing research eligibility; no rich environment or commercial reward
GBrain / file wiki	Public memory implementations, provenance and retrieval	Pinned component comparison; no whole-product ranking

Table 1: Nearest precedents. Their reported scores use different tasks and cannot be placed on our leaderboard.

2 Position relative to prior work

Our search covers agent memory, domain specialization, temporal/provenance handling, memory-to-action, marketing, evaluation and classical reasoning. The initial bounded search retained 399 unique arXiv records; an additional six-query search returned 113 unique identifiers, with overlap between searches. These are discovery counts, not full-text review counts. Dated queries, screening depth and nearest-work notes are released. Citation metadata is verified against owning sources; the latest search cutoff is 22 October 2026 (UTC). Access and coverage limits remain explicit.

RAMP is the closest domain precedent. With GPT-4.1, its Table 1 reports audience accuracy 0.583 for actor-only and 0.870 with planner plus semantic memory; the combined intervention cannot isolate representation. Adding verification/reflection yields 0.853 in Table 2, illustrating that extra machinery need not help. Its 88 queries and small product-manager study are valuable evidence for audience curation, distinct from company-held-out memory specialization. We do not claim to introduce marketing memory plus planning.

Contract Memory Compiler (CMC) is a strong methodological precedent: freeze the reader and extraction, change state resolution/retrieval, and include length controls [2]. Its last-write-wins assertions differ from approved policy and unresolved disagreement. WiCER demonstrates that blind wiki compilation can lose facts and that raw full-context versus RAG direction changes across corpora [3]. We therefore retain full-context references and a separate compilation diagnostic. Neither a graph nor Markdown alone defines specialization. Domain memory, fact/belief separation and recommendation have further precedents in IFCMemoryBench, Hindsight and belief-state recommendation [6–8]; case-based reasoning and truth maintenance predate modern agents [9, 10].

Broader agent-memory surveys, temporal graph memory, GraphRAG and production-oriented memory systems span different storage, extraction and retrieval choices [11–14]. We do not rank these unexecuted systems against our selectors. The released review matrix distinguishes metadata screening, full-text inspection, artifact inspection and reproduced components; reported upstream headline scores remain separate from our measurements.

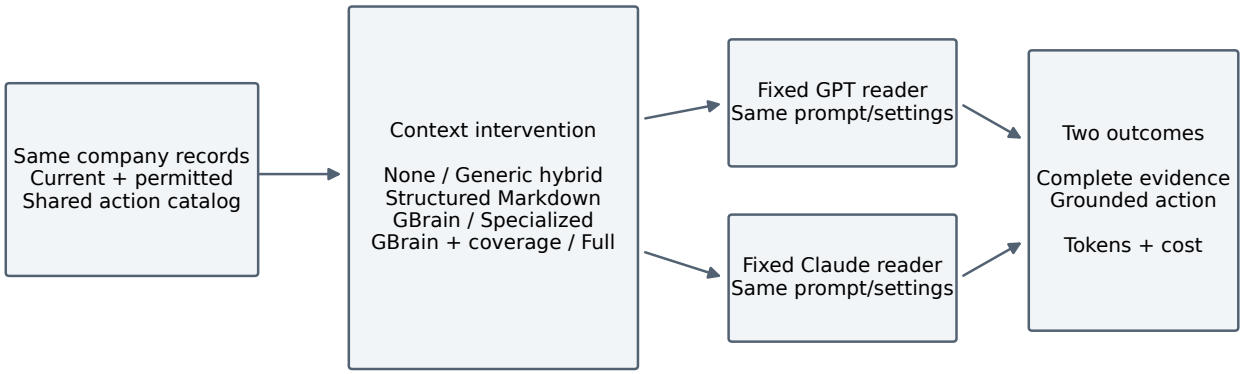


Figure 1: Shared sources and task, changed evidence packet, fixed reader. Context completeness is measured before the reader; action groundedness is measured afterward. The pipeline does not imply commercial causality.

3 A controlled context-to-action experiment

3.1 Data, boundaries and task

We instantiate 33 fresh fictional company worlds, 264 queries, across renewal, partner and compliance families. Four development worlds use hiring and migration. The new main worlds are disjoint from both the pilot and earlier experiments; the relation families were used in historical development and are not previously unseen research concepts. Each world contains a seller, a target account, a shared four-action catalog and 12–36 distractor records. Four conditions are answerable and four require withholding. Conditions span initial context, weak feedback, policy revision, account requalification, evidence correction, restricted evidence, objective change and policy conflict.

Every method receives the same verified events with source ID, valid time, observed time, authority and access scope. A shared visibility/supersession function removes future, superseded and unauthorized records before selection. Consequently, the experiment measures *selection after common maintenance*; it does not establish that one memory discovers updates or permissions better. Weak feedback cannot become approved policy. Conflicting current values remain explicit. The data are original, AI-assisted templates with deterministic construction labels cross-checked by Python and SQLite implementations, not customer observations or human annotations.

A reader returns at most one action, source bindings, missing information and a short explanation. An action must match the objective and resource allowance and satisfy its own prerequisites and cross-field equalities. Multiple catalog actions may be acceptable. The public catalog supplies requirements to every arm; hidden labels and future events never enter requests. Schema validity, evidence grounding and prerequisite satisfaction are machine-checkable. A separate post-hoc mock executor checks internal workflow consistency, not outreach effectiveness; the scorer's catalog-feasibility receipt alone is not execution.

3.2 Seven contexts, two fixed readers

BM25 and reciprocal-rank fusion follow standard retrieval practice [15, 16]. All dense retrieval uses actual `text-embedding-3-small` vectors with 1,024 dimensions. GBrain[1] is pinned to `531b3b1` (version 0.60.380.0); its semantic path is verified in saved receipts. We exercise native search over common

[1] https://github.com/garrytan/gbrain; MIT-licensed upstream, installed separately.

Treatment	Executed mechanism
No company context	Shared task/catalog only; information-availability reference
Generic hybrid	BM25 plus real dense embeddings; reciprocal-rank fusion ($k = 60$)
Structured Markdown	Lossless entity files; generic ranking diversified by relation/authority
GBrain search	Actual pinned PGLite balanced hybrid search, same real vectors
Specialized coverage	Generic ranking prioritized by public action prerequisites, then diversified
GBrain + coverage	The identical prerequisite-coverage layer over native GBrain candidates
Full current context	Every permitted seller/account record; larger-resource reference

Table 2: Only context selection changes within a reader. All arms share a strong domain-aware prompt; Generic is therefore also the prompt-specialized control.

atomic records, with reranking off and no extracted source graph, synthesis or native ingestion. This is a component evaluation, not GBrain's best complete configuration.

The specialized layer prioritizes relation/authority groups required by feasible catalog actions. It never filters by the correct value or reads a label. Generic file diversity covers relation/authority groups without catalog priorities, making it an essential falsifier: generic coverage may suffice. The controlled Markdown representation is lossless and deterministic, not an LLM-generated wiki or a claimed Karpathy implementation. Section 5 separately evaluates actual wiki compilation.

Primary evidence allowance is 2,048 UTF-8 bytes in the same canonical record encoding. Instructions and catalog are common and excluded from that allowance. Whole records are packed without truncating their contents. This is an equal serialization budget, *not* exact tokenizer equality; actual provider tokens and costs are reported. Retrieval-only sweeps use 1,024, 2,048, 4,096 and 8,192 bytes. Full context and no context are explicitly unequal-information references.

The readers are `gpt-6.1-sol` (low reasoning) and `claude-sonnet-5-5` (provider defaults), each with 2,048 maximum output tokens and the same public JSON schema. Within each provider, settings and prompts are identical across arms. There is one request per case/arm/provider, shuffled deterministically; no answer retries or external tools. A Claude/GPT difference would also include different provider defaults and tokenizers, so our primary contrasts are *within* each model.

3.3 Outcomes, precision and accounting

Context quality is the proportion of answerable queries whose packet contains at least one complete current support set. Grounded decision correctness counts a supported eligible recommendation on answerable cases and correct withholding on negative cases. We separately report positive coverage, inappropriate abstention, unsupported recommendations, false apparent eligibility, failures and a deterministic validator's effect. The validator cannot import missing facts or repair citations. Scores are recomputed from raw saved responses without an LLM judge.

The completed pilot issued 448 requests. Its paired world-level SD selected 33 main worlds through the prespecified rule $n = \lceil (1.96s/0.075)^2 \rceil$, rounded to a multiple of three and bounded 24–60. The main seed, code, input hashes and analysis were frozen before main requests. No resizing or outcome-based stopping follows. Paired differences use world means; 95% bootstrap intervals resample worlds within family (10,000 draws). Exact paired sign tests for Specialized–Generic are Holm-adjusted across the two readers. Other contrasts are descriptive. Clustering does not turn templated worlds into real-company samples; degenerate intervals at a ceiling reflect the observed fixtures.

Costs use actual reported input, output and cache usage and official prices checked 22 October 2026. Both readers list \$2/M uncached input and \$10/M output; cache reads/writes are costed separately. Estimates are not reconciled invoices. Ingestion, queries and compilation are separated. All attempts, including the stopped serialization pilot and transport failures, remain available. Researcher-selected execution guards bound requests; they are not user-specified spending limits.

Context	Complete support	GPT correct	Claude correct
No context	0/132	132/264	132/264
Generic hybrid	91/132	223/264	223/264
Structured Markdown	132/132	264/264	264/264
GBrain search	20/132	152/264	152/264
Specialized coverage	132/132	264/264	264/264
GBrain + coverage	132/132	264/264	264/264
Full current context	132/132	264/264	264/264

Table 3: Frozen main study at 2,048 evidence bytes. Complete support is measured on 132 answerable cases; grounded correctness includes those 132 cases and 132 required-withholding cases. Each reader executes 264 requests per row. Full context is an unequal-budget reference.

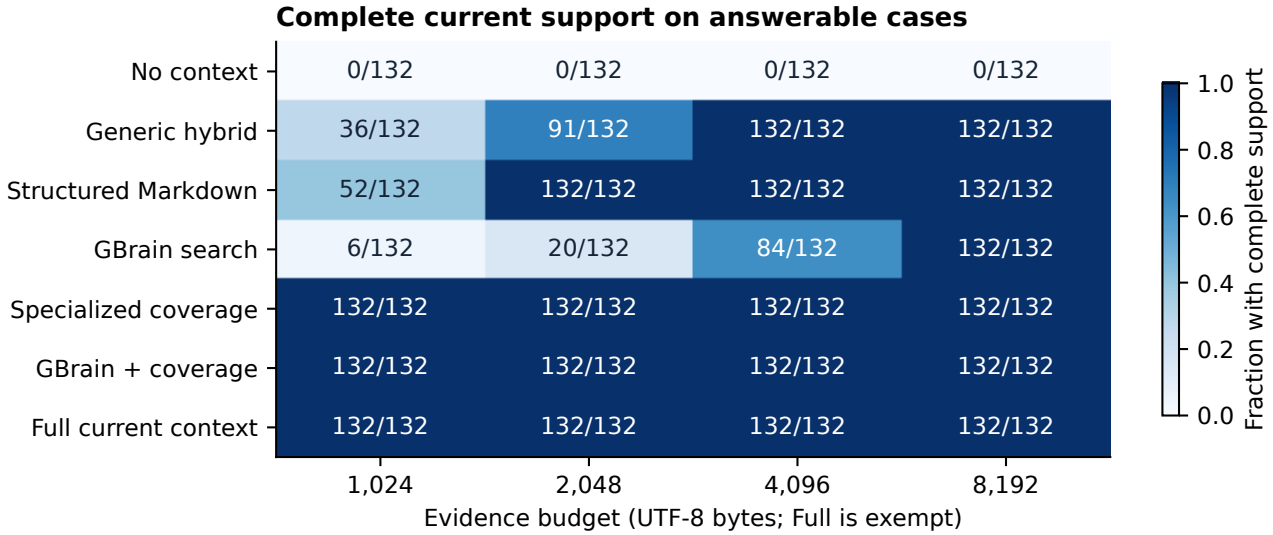


Figure 2: Retrieval-only budget sweep on the same 132 answerable cases. At 1,024 bytes, specialization preserves complete support while generic diversity often cannot. Generic hybrid closes the gap at 4,096 bytes; native GBrain search does so at 8,192. Reader experiments use 2,048 bytes only; lower-budget recommendation gains are not inferred from this figure.

4 Main results: coverage explains the observed action gain

All 3 696 planned requests completed, with no schema or transport failures. The audit reconstructs 7,392 context packets and re-scores every raw response. The same-backbone results are identical for the two readers on this formal task; that agreement is not two independent company samples.

Context availability. Generic hybrid retrieves complete support for 91/132 answerable cases (68.9%); specialized coverage retrieves 132/132. Native GBrain search supplies 20/132 under this small allowance, while adding the same coverage layer raises it to 132/132. Structured Markdown also supplies 132/132 without catalog-specific priorities. This last control rules out a claim that a domain ontology or proprietary backend is required at the primary budget.

Grounded decisions. Specialized context improves each reader from 223/264 to 264/264 correct decisions (Figure 3). The paired gain is 15.5 percentage points, 95% interval [9.8,21.2]; the Holm-adjusted exact paired sign-test value is 1.22×10^{-4} for each provider. The effect is positive in 15 worlds and zero

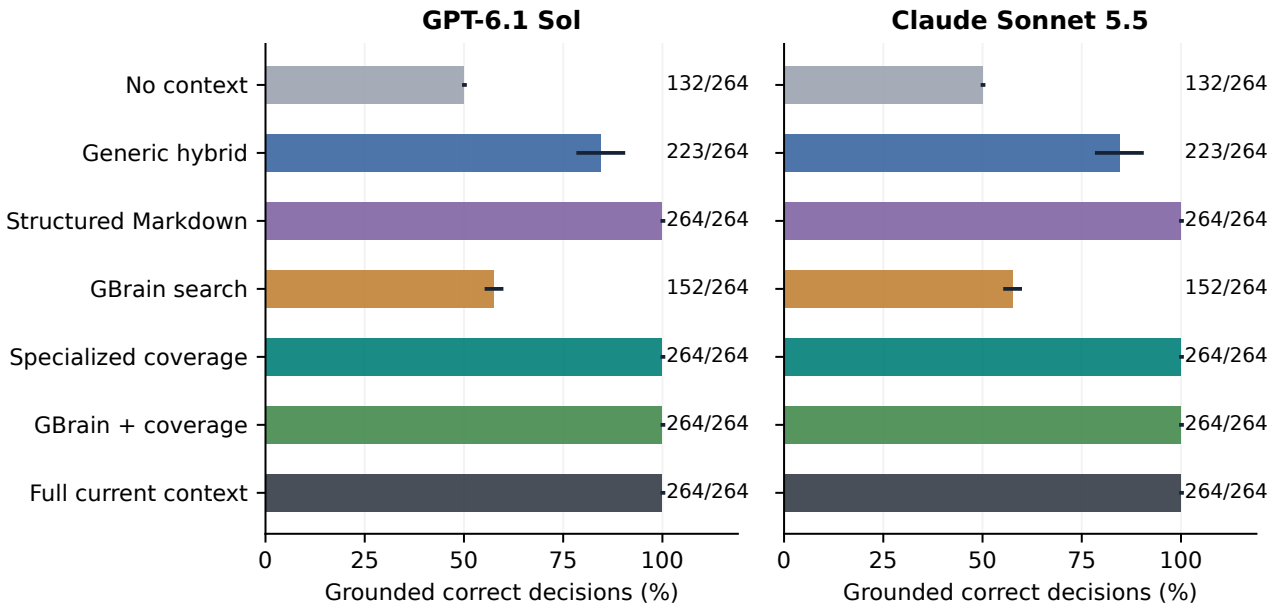


Figure 3: Grounded correct decisions with fixed readers. Bars start at zero; lines are 95% family-stratified world-bootstrap intervals. Several methods reach the fixture ceiling, so their empirical intervals collapse; this does not establish universal perfection.

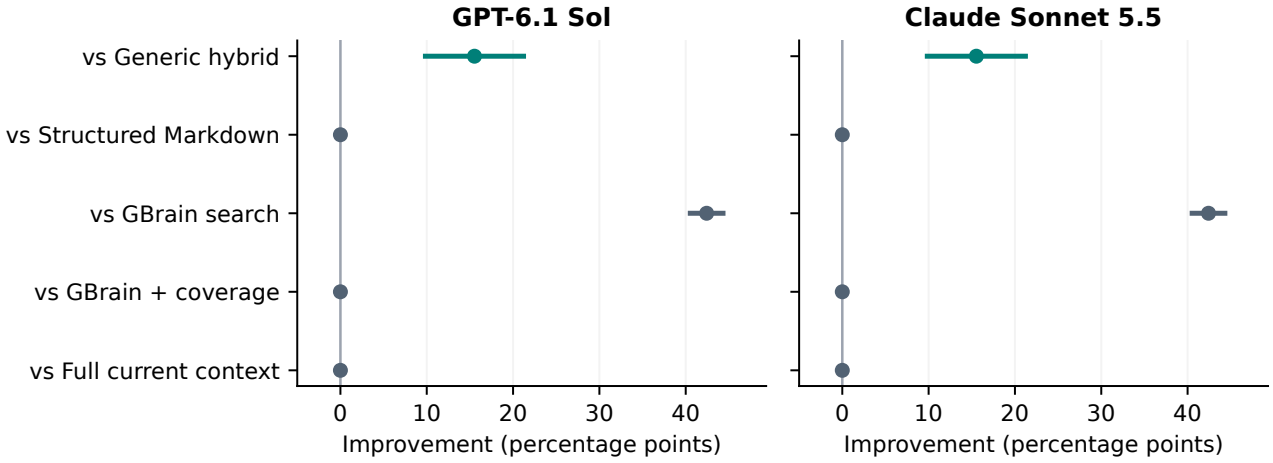


Figure 4: Paired Specialized-minus-comparator differences. The primary Generic contrast is highlighted; the other comparisons are descriptive. Zero differences against Markdown, GBrain+coverage and Full are observed ties, not formal equivalence tests.

in 18. Positive recommendation coverage rises 31.1 points, from 68.9% to 100%: 41 additional supported recommendations among 132 answerable cases, or 45.1% more than the 91 produced with Generic. These percentages describe formal eligibility and citations, not a 45.1% improvement in strategic value.

The gain is recovered coverage, not reduced hallucination. Every arm withholds on all 132 negative cases per reader. No unsupported recommendation is observed. Generic's 41 missed positives and GBrain search's 112 missed positives per reader are abstentions where complete evidence was absent. Every positive packet with complete support produces a grounded action. The common validator

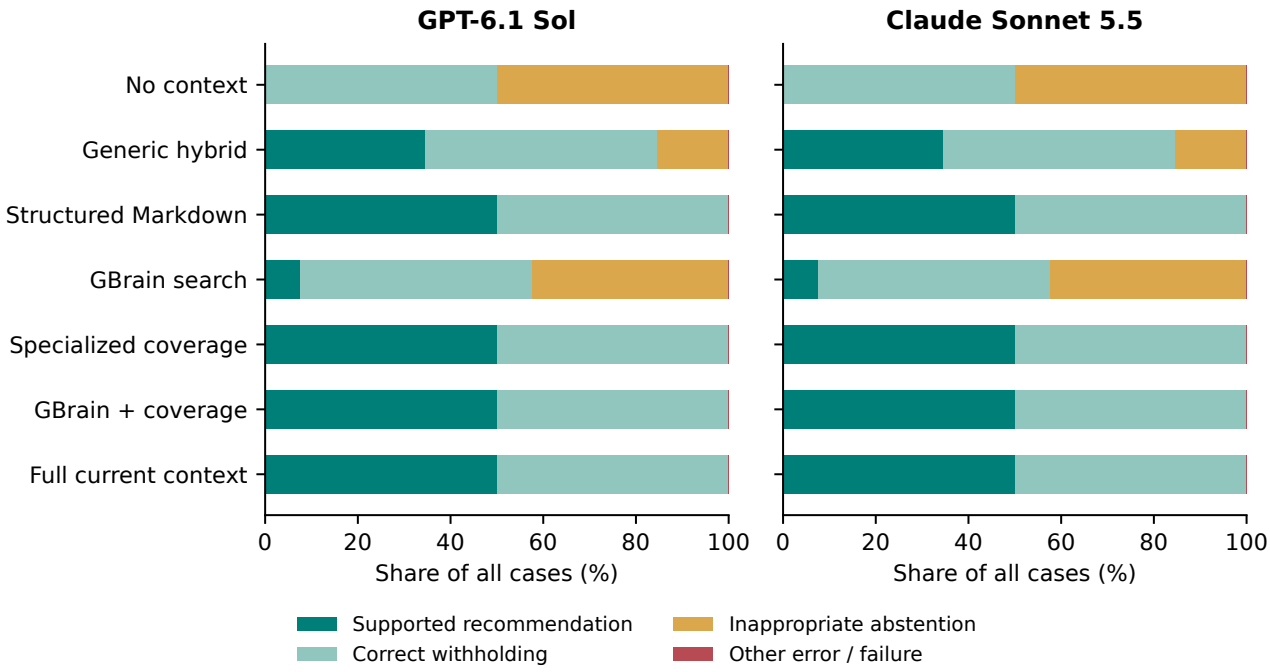


Figure 5: Outcome composition over all 264 cases per reader/arm. Supported recommendations and correct withholding are distinct. The remaining mass is inappropriate abstention on answerable cases; observed unsupported/error mass is zero.

changes no decision. No-context correctly withholds but covers none of the positive cases: its 50% overall accuracy is an information-availability reference, not evidence of a weak model.

The qualitative audit selects the first Generic-failure/Specialized-success case in each family by sorted case ID. Missing seller geography blocks the partner and compliance cases; missing approved role, geography and offering pain block the renewal case. The same GPT correctly requests evidence, then recommends when coverage supplies those fields. Structured Markdown supplies them too. These traces localize the failure to evidence selection and show why a small general coverage mechanism can match specialization.

Resources and action-interface checks. The main reader calls cost an estimated $33.99 from recorded usage. Specialized input averages 2,032 GPT tokens and 3,966 Claude tokens, versus 2,977 and 5,536 for Full: 31.7% and 28.4% less input with the same observed decisions. Total price also depends on output and cache behavior; it is not a pure measure of retrieval efficiency. Table 5 reports actual batch operating points. Cold local GBrain setup, source curation and production maintenance are not assigned invented dollar costs.

A separate post-hoc in-memory execution check completes all 1,278 proposed workflows through 2,606 mocked capability calls, with no external actions. It reads only supplied observations, verifies cited prerequisites and generates an evidence report within the declared allowance. This checks interface consistency; the scorer's simpler feasibility receipt is not mislabeled as execution, and neither check establishes campaign effectiveness.

The exact SQL reference solves all 264 cases from the permitted verified records without model calls. An LLM is therefore unnecessary for this formal eligibility problem once the facts and rules are clean; the comparison isolates evidence delivery to language-model readers, not a need for generative decision logic.

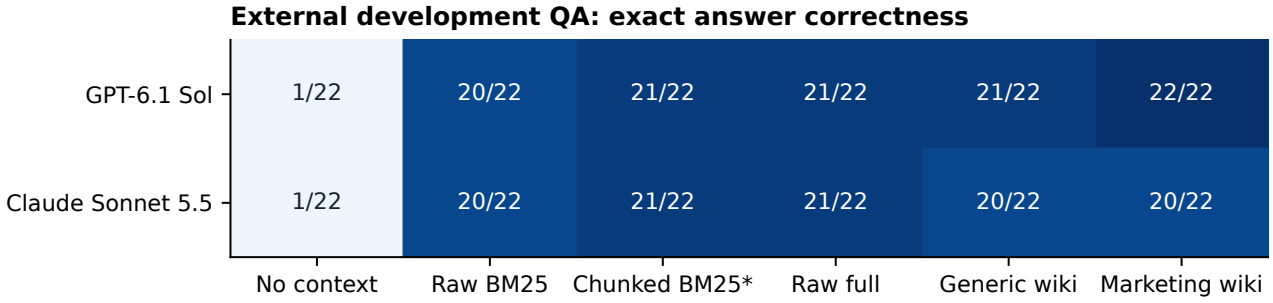


Figure 6: External development QA, 22 questions per cell. The starred chunked-BM25 control was added after error inspection; all original scores remain. Unequal context lengths, one company and visible development labels prevent a confirmatory specialization claim.

5 External documents: does compiling a wiki help?

A second, exploratory track uses 118 MIT-licensed development pages from GBrain's externally authored `messy-company-v1` corpus: one fictional company, eight account histories and 22 supplied questions. The research agent inspected the development labels; this is not a hidden test or independent expert validation. It asks factual retrospective QA, not action choice. The snapshot grants access to all documents, including fictional private-marked notes. It does not test causal observation-time or permission enforcement.

We compile the same raw documents with GPT into a generic company wiki and a wiki organized around marketing decisions. The compiler receives no questions, labels or answer feedback. Both use low reasoning, the same 32,000-token maximum and identical preservation instructions apart from organizational schema. The final wikis contain 66,742 and 63,942 bytes, respectively, versus 293,342 bytes for the rendered raw source packet: reductions of 77.2% and 78.2%. These are original, single-pass Markdown compilations inspired by file-memory practice, not an official Karpathy system.[2] Native asynchronous generation resolves transport limits; previous timed-out and truncated attempts are retained.

Figure 6 compares no context, whole-document BM25, full raw documents and both compiled wikis with each fixed reader. The marketing wiki scores 22/22 with GPT and 20/22 with Claude; the generic wiki scores 21/22 and 20/22. Full raw context scores 21/22 with both. All 220 original reader calls completed. Every returned citation ID is valid, but this check does not establish that a cited passage entails the answer.

A stronger simple retrieval control. Whole-document BM25 scores 20/22 for both readers. Inspection found that the 58,320-byte master agreement could never fit its 24,576-byte allowance, regardless of rank. An explicitly post-hoc 44-call control uses 2,048-character windows with 256-character overlap, the same BM25 and byte allowance, and no label-based filtering. It scores 21/22 for each reader and restores the missing 120-day clause. Chunked BM25 costs $0.370 for 22 GPT calls and $0.509 for 22 Claude calls, compared with $0.905 and $1.216 for the marketing wiki's reads, before compilation. Thus a wiki's compression relative to full context does not imply an advantage over competent chunked retrieval.

The last case in Table 4 is consequential. A disclosed post-hoc sensitivity removes that one question without changing any primary label: GPT's marketing wiki, generic wiki, full context and chunked BM25 all score 21/21. Claude scores 19/21 with the marketing wiki, 20/21 with the generic wiki, and

[2]We inspected the unofficial MIT implementation at https://github.com/Astro-Han/karpathy-llm-wiki. Karpathy's owning Gist was inaccessible from this runtime; we do not quote it or claim to execute an official release.

Observed case	Source/answer diagnosis	Implication
Notice period	Whole-document retrieval omits a long agreement; both readers correctly say unknown. Chunking recovers the 120-day clause.	An ingestion/packing defect can masquerade as a memory-quality gap.
Dated account alias	The raw September 15 update explicitly places "Dana's freight account" under Quarrendon. Compiled wikis omit that phrase; Claude misidentifies or withholds.	Preserving entity lists need not preserve a contextual relation.
Buyer-role change	March CRM names the CFO; May follow-up distinguishes COO decision maker from budget approver. Claude treats the marketing wiki's preserved claims as unresolved.	Evidence preservation and usable resolution are different; reader behavior matters.
Ambiguous reference	July prose mentions a Quarrendon visit, but Dana holds two freight accounts. Upstream requires "ambiguous"; both full-context readers infer Quarrendon.	One plausible-reference disagreement can determine a small benchmark's ranking.

Table 4: Qualitative inspection of observed failures by the same AI-assisted research process, not blinded human review. Source chains and raw explanations are released.

21/21 with full or chunked raw context. The apparent one-answer GPT advantage is therefore not robust evidence of general marketing-wiki superiority. No wiki is refined in response to these errors.

6 What the evidence supports

The engineering implication is concrete: preserve enough distinct, authoritative evidence to satisfy an action's prerequisites before asking a model to decide. With competent readers and bounded context, that changes usable recommendation coverage. It can be implemented as a small selection layer over a general memory system; the successful GBrain+coverage control makes replacement of the storage backend unnecessary in this experiment.

The useful specialization is also conditional. Catalog priorities matter under the tightest retrieval allowance, but ordinary relation diversity matches them at 2,048 bytes, and more context erases the remaining retrieval gap. Full context is a serious option for a small company knowledge base. On messy external prose, compilation loses a dated relation and can preserve conflicting claims in a form that one reader overinterprets as unresolved. A stronger chunked baseline eliminates an apparent retrieval defect. A specialized schema is therefore an engineering hypothesis to evaluate, not a universal performance guarantee.

For product design, the evidence favors versioned sources, explicit authority and scope, complete prerequisite coverage, and visible requests for missing evidence. It supports benchmarking both the memory output and each downstream reader. It does not support a claim of a proprietary graph advantage, a learned revenue policy or a network effect. Accumulated company context may become valuable through maintained, reusable evidence; customer retention and economic defensibility require separate evidence.

Limits of inference. The main task is small, synthetic and rule-specified. It assumes verified events and a shared current-view filter, uses known relation families and a four-action catalog, and reaches a ceiling with strong readers. It does not evaluate raw extraction, independent update maintenance, native ACLs, large heterogeneous company memory, adversarial instructions or open-ended strategic creativity. One response per treatment estimates behavior at the recorded settings, not model-wide

reliability. Bootstrap intervals condition on generated worlds and a fixed family mix. Byte equality is not tokenizer equality or full lifecycle cost equality. API model aliases may change; exact returned identifiers and traces are retained.

The external corpus reduces same-project authorship but remains AI-authored development material from one fictional seller. There is no blinded human usefulness study or independent external reproduction. Qualitative inspection is performed by the same AI-assisted research process and explicitly labeled. The compiled wiki, native search and structured files each exercise only part of a complete memory product. All failed attempts and post-hoc controls are disclosed. Ready-to-release code and a compiled report are separate from public distribution, peer review or paper submission.

7 Conclusion

At matched evidence allowances, better coverage can improve grounded marketing action recommendations without changing GPT or Claude. In the controlled study, the gain over general hybrid retrieval is 15.5 percentage points for both readers. Strong generic coverage and a composable GBrain extension match the specialized system at the primary budget; tighter retrieval budgets reveal where domain priorities add value. The practical choice is to measure and maintain action-relevant evidence coverage, while testing the downstream reader and retaining simple full-context and file-memory alternatives.

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A Protocol, implementation and reproduction details

The main freeze is `experiments/integrated/main-freeze-v1.json`, committed at `61c57c6`. Seed 2026102302 produces 33 worlds (11 per family) and 264 cases. The input hash starts `449072b7255f`. Pilot seed 2026102231 produces four worlds and 32 cases. The pilot's maximum paired world SD was 0.2165, selecting 33 worlds through the prespecified precision rule. Earlier test families informed development; this is a fresh-world evaluation within a known generator, not a claim of wholly unseen task design. Each case is executed once per context/provider, with native structured output and no answer sampling. Software source snapshots and complete hashes accompany the run.

GBrain's full pin is `531b3b1ab66c88f192abb51338028c44b80fbf21`. The external development corpus comes from `garrytan/gbrain-evals` at `6999ebc9da0ac784ac05995b586d22293544e924`; per-file URLs and hashes are retained. These repositories and the original adapters have MIT licensing; original generated fixtures/documentation have CC BY 4.0 scope. Third-party paper PDFs, copied abstracts, credentials, production code and private planning history are excluded from the public package. No GBrain hosted configuration or official Karpathy implementation is claimed.

The common event filter is deliberately shared. It performs observed/valid-time gating, scope checks and explicit supersession. "History handling" in this study therefore means selecting from a maintained current view; no independent advantage in ingestion, belief revision, authorization or update latency is estimated. Source aliases are generated from the supplied packet, while all permissible action IDs remain in the public schema. The JSON schema fixes serialization, not the correct decision. No-context contains no source aliases and cannot invent evidence.

Failed development attempts. The initial provider pilot was stopped after 132 issued requests: 124 completed and eight Claude responses had malformed JSON. Its remaining 316 plans were not issued. The amended native-schema pilot completed 448 requests. Both attempts remain, with no successful replacement inserted into the failed ledger. Schema-integration errors, local embedding-path failures and synchronous wiki failures are also preserved. Reported usage excludes unknown billing when a timeout prevented a usage receipt; unknown is not zero. During the main run, a runtime reconnection stopped the process after eight response/result files were saved but before their terminal ledger entries. Those exact verified files were recovered, and only the remaining 624 unissued requests were resumed. No generation was repeated; the before/after journals and recovery receipt are retained.

Reproduction scopes. The inexpensive smoke tests use no provider calls. The saved-output path rebuilds data, every context packet, raw-response scores, statistical summaries and vector figures, then compiles both PDFs. A clean extracted directory and isolated, hash-pinned Python environment test this path; the hardware and system TeX are shared with the original run. Fresh inference is a separately priced command requiring current provider access; it may not be byte-identical. Fresh native GBrain search requires the pinned upstream/Bun dependencies. Fresh embeddings are optional when using the released actual vector snapshot. Commands and requirements appear in the companion README.

Historical evidence. The previous local Qwen3-30B-A3B study and exploratory GBrain extension are archived rather than pooled into the new sample. They used different readers, budgets and data. They motivated competent reader checks, a full-context reference, wider-budget sweeps and generic coverage controls. Exact historical PDFs, attempts, source hashes and reproducibility receipts remain under `release/history/` and the original experiment directories. The new paired effects come exclusively from the frozen integrated main study.

B Measured reader operating points

Across provider pilots, compilation and readers, 4,552 unique saved generation responses and 12 embedding responses have known usage totaling an estimated $55.49, including $0.000385 for embeddings. Three unsuccessful attempts have no usage receipt and may have incurred additional charges. The main reader subtotal is $33.99. Hardware, curation, credits and taxes are excluded; these are list-price estimates, not invoices or total ownership costs.

Reader	Context	Input	Output	$/100 calls	Seconds
GPT-6.1 Sol	No context	808	224	0.385	5.06
	Generic hybrid	2027	202	0.474	4.89
	Structured Markdown	2031	198	0.358	4.89
	GBrain search	1963	223	0.532	5.14
	Specialized coverage	2032	199	0.337	4.89
	GBrain + coverage	2026	200	0.419	4.88
	Full current context	2977	203	0.756	4.95
Claude Sonnet 5.5	No context	1711	341	0.683	3.23
	Generic hybrid	3950	630	1.420	5.71
	Structured Markdown	3964	690	1.482	6.00
	GBrain search	3842	446	1.215	4.30
	Specialized coverage	3966	697	1.490	6.04
	GBrain + coverage	3955	684	1.475	5.95
	Full current context	5536	740	1.847	6.99

Table 5: Mean input/output tokens per request, realized usage-based price per 100 calls and mean response time in the main batch. Provider tokenizers/defaults differ. Cache behavior, bounded concurrency and the runtime interruption affect operating points; these are not service-level or production-TCO estimates. Source/embedding and compilation costs are reported separately.